

Minsun Park

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Data Scientist - Recommendation, Ranking, Experimentation & ML Systems

SUMMARY

Data scientist with 5 years of experience building and improving recommendation, ranking, and ML systems through large-scale analysis, controlled experiments, and production ML. I investigate performance changes across data, models, exposure logic, and system behavior; turn ambiguous findings into testable hypotheses; and work with product and engineering partners to ship practical improvements. My work spans behavioral data design, retrieval and ranking, statistical validation, online experiments, and reliable ML pipelines.

EXPERIENCE

Data Scientist, Kakao Style - AI Search & Recommendation Sep 2024 - Present

- Designed and analyzed a **5-arm A/B test** for personalized recommendation blending; traced the missed target across model, data, and serving behavior to exposure shuffling that discarded the computed rank. The resulting dataset and API were adopted in the production personalized banner-ranking score.
- Built the company's first store-level trait system (**23,000 stores x 70+ features**) and a similar-store recommender combining text, SVD, and Fashion-CLIP embeddings; translated qualitative evaluation and error analysis into ranking heuristics, shipped the system to search personalization, and **cut batch runtime 80%**.
- Developed a user size prediction framework **serving millions of users daily**; evaluated three inference paths on 10M+ purchase records, established a two-factor confidence policy, and increased recommendation coverage from **82.6% to 92.1%** while maintaining approximately 91% top-3 accuracy.
- Improved the efficiency and reliability of a production recommender by migrating SageMaker training to LightGCN on MLflow, **reducing training time 74%** and **embedding API cost 67%**; diagnosed DAG OOM failures, validated KServe serving, and integrated real-time Kafka attributes.

Data Scientist, Kolon Industries FnC Sep 2021 - Sep 2024

- Built **CTR-prediction-based personalized recommendations** with topic-modeling re-ranking and behavioral features for commerce surfaces; partnered with business teams to translate engagement patterns into targeting and merchandising decisions.
- Developed product-to-query retrieval models (BERT, DPR, ColBERT) for advertising keyword extraction and a related-search recommender whose launch reduced search-page bounce rate from **10.9% to 5.4%**.
- Built demand-forecasting models that halved reorder lead time (**8 to 4 weeks**) and an SKU optimization model deployed company-wide (**16% fewer SKUs**); communicated trade-offs and recommendations to operational stakeholders and received an internal excellence award.

Statistical Analyst Intern, Bank of Korea Feb 2019 - Sep 2019

- Validated national balance-of-payments statistics (900+ line items) in SAS and R, identifying misreported transactions whose corrections were incorporated into official statistics.

Data Analyst Intern, Early Adopter (advertising agency) Oct 2016 - Jan 2017

- Analyzed user behavior and conversion funnels and ran an ad-channel A/B test; findings informed the next year's channel strategy, while a data-driven content proposal generated **1.34x** the prior month's views.

EDUCATION

M.S., Applied Artificial Intelligence - Korea University (expected Feb 2027) Sep 2024 - Present

B.S., Applied Statistics - Chung-Ang University Mar 2015 - Aug 2021

SKILLS

Applied Science: Experiment design, statistical validation, hypothesis-driven root-cause analysis, recommendation & ranking, forecasting, optimization, error analysis

Languages & Data: Python, SQL, PySpark, Spark, SAS, R

ML & Systems: PyTorch, TensorFlow, scikit-learn, MLflow, OpenSearch (kNN), Airflow, Databricks, Kafka, Kubernetes (KServe)

Cloud: GCP (BigQuery, GCS) · AWS (SageMaker, Bedrock, EMR, Athena, S3)

Languages: Korean (native) · English (fluent) · TOEIC 960, TOEIC Speaking Lv.7

HONORS & AWARDS

- 1st Place, AWS AI Agent Workshop** (2026) - designed an agent that recommends sizes to reduce returns
- Man of Kolon Award** (2024) - company-wide recognition for an SKU optimization model deployed to production
- 1st Place, SSAFY 6th Hackathon** (2021) - destination recommendation using public bike-share histories

RESEARCH & KNOWLEDGE SHARING

- Thesis in progress: **filtered approximate nearest neighbor search over HNSW indexes**, analyzing the trade-off between pre-filter path fragmentation and post-filter recall loss
- Run a weekly research radar on recommendation, ranking, and NLP papers and translate relevant findings into practical team recommendations
- Proposed and seeded a team-wide practice of documenting AI failure patterns into shared engineering guidelines